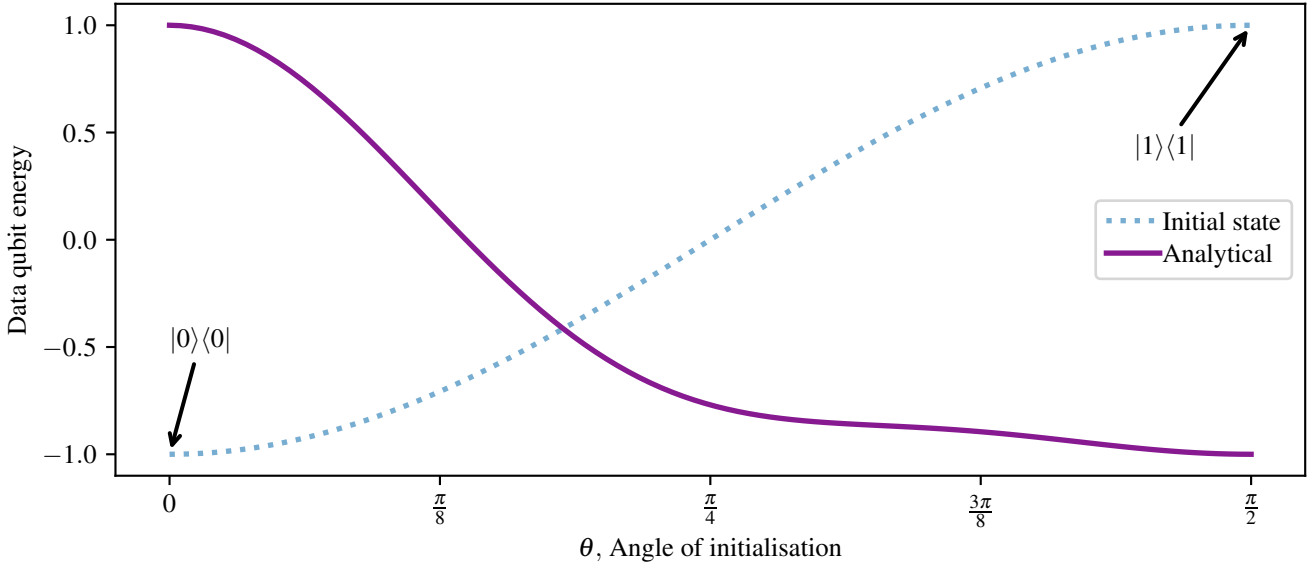

1.0
0.5
0.0
−0.5
−1.0
Data qubit energy
0
$\frac{\pi}{8}$
$\frac{\pi}{4}$
$\frac{3\pi}{8}$
$\frac{\pi}{2}$
θ, Angle of initialisation
$|0\rangle\langle 0|$
$|1\rangle\langle 1|$
Initial state
Analytical